# TECHNICAL EVALUATION — SUBMERSIBLE PUMP

## 1. Evaluation output

| Field | Value |
|---|---|
| **Report** | submersible_pump_EN201222034S.pdf (89 pages) |
| **Standards** | IEC 60335-1:2010 (Part 1); IEC 60335-2-41:2021 (Part 2 2-41) |
| **Model** | no-llm (markdown) |
| **Created** | 2026-11-01 08:33 UTC |
| **Evaluation reference** | 81DC43D1 |

## 2. Report identity

| Field | Value |
|---|---|
| **Report number** | EN201222034S |
| **Laboratory** | EMTEK(NINGBO) CO., LTD. |
| **Report date** | Dec 28, 2025 |
| **Product declared** | Submersible pump |
| **Declared model** | SPC55068 |
| **Standards claimed** | IEC 60335-1:2010+A1:2013+A2:2016; IEC 60335-2-41:2025 on page 2; IEC 60335-2-41:2012 on page 3 |
| **Standards evaluated** | US IEC 60335-1:2010; IEC 60335-2-41:2012 as reproduced in ILNAS-EN IEC 60335-2-41:2021 |

## 3. Summary

| Field | Value |
|---|---|
| **Final verdict** | REPORT CORRECTION / REPLACEMENT REQUIRED — PRODUCT COMPLIANCE NOT DETERMINED — CERTIFICATION HOLD |
| **Reason** | The tested model and electrical test basis conflict with the declared product, and several applicable safety assessments lack substantiated evidence. The claimed 2025 particular part and Part 1 amendments were not supplied, so those claims cannot be evaluated. |
| **Recommended action** | not stated |
| **Conclusion** | The report does not demonstrate compliance of SPC55068 with the supplied general and particular requirements. A replacement report must establish sample identity and ratings, resolve the conflicting test conditions, and provide the missing marking, immersion, cord, earthing, insulation and heat-and-fire evidence; no product non-conformity is established solely from the mismatched measurements. |
| **Findings** | 22 total — Critical 4, High 16, Medium 2, Low 0 |
| **Evidence check** | Report quotes found 19/22; standard quotes found 20/22; not found (needs review) 0 |

## 4. Findings

'(code)' = the values in this row were also checked by code against the report. Evidence = the AI's verbatim quotes searched by code in the report and standard PDFs; NOT FOUND means the row needs human review.

| Priority | Clause | Gap type | Normative requirement | Finding reason | Required action | Pages | Evidence |
|---|---|---|---|---|---|---|---|
| **Critical** | 5.2; 7.1 | Not Stated | The test evidence cannot be reliably attributed to the declared model. | The declared model is SPC55068, but the summary states that full tests were performed on model 4TSC4.0-96-110/1000. No relationship between the models is established. | Provide a replacement report identifying the tested sample and establishing its relationship to SPC55068. | PDF pp.2–3 | Report: verified p.3 Standard: verified in IEC 60335-1:2010 p.26 |
| **Critical** | 7.1 (code) | Not Stated | The voltage used to identify the appliance and assess its marking is inconsistent with the declared rating. | The declared rating is 220–240 V, but the marking assessment records 96 V. | Verify the actual marking and correct the report to identify the applicable rated voltage. | PDF pp.2–3,6 | Report: no quote given Standard: verified in IEC 60335-1:2010 p.29 |
| **Critical** | 10.1 (code) | Not Stated | The reported measurement is not evaluated against the declared model's rating and rated voltage. | The declared rated input is 550 W, while the power-input table uses 1,000 W and records 995.2 W at 72 V. | Test and report the power input of SPC55068 against its verified rating. | PDF pp.2–3,79 | Report: no quote given Standard: verified in IEC 60335-1:2010 p.38 |
| **Critical** | 5.8.1; 5.8.2; 11.5; 13.1; 16.2 | Not Stated | The recorded supply and test voltage do not establish testing at the applicable conditions for the declared rating. | The report declares 220–240 V, 50 Hz, but identifies the supply as DC and records 76.3 V for heating and leakage-current tests. | Verify the supply type and repeat or substantiate the affected tests at the applicable voltage and frequency. | PDF pp.2–4,14–15,79–80 | Report: no quote given Standard: verified in IEC 60335-1:2010 p.27 |
| **High** | 5.101; 25.14 | Not Stated | Pumps are tested as portable unless fixed; the particular part subjects portable pumps, except specified types, to the flexing test. | Portable and fixed classifications are not identified in the particulars; testing as a portable pump and the cord-flexing test are marked N/A. | Establish the pump classification and perform the applicable cord-flexing test. | PDF pp.4,6,43 | Report: verified p.6 Standard: verified in IEC 60335-2-41:2021 p.13 |
| **High** | 7.1 | Not Stated | A submersible pump above 50 W must be marked with its maximum operating depth in metres. | Maximum operating depth is recorded as N/A for the declared 550 W submersible pump, and no depth is shown on the reproduced marking plate. | Verify and document the maximum operating depth marking on the tested pump. | PDF pp.2–4,7 | Report: verified p.4 Standard: verified in IEC 60335-2-41:2021 p.14 |
| **High** | 7.1; 5.7; 11.7 | Not Stated | The pump must be marked with a maximum liquid temperature; liquid temperature determines the specified test conditions. | The summary gives maximum liquid temperature as "<45□", but the reproduced marking plate gives no maximum liquid temperature and no actual liquid test temperature is recorded. | Verify the temperature marking and document the actual liquid temperatures used in testing. | PDF pp.3,6–7,14,79 | Report: verified p.3 Standard: verified in IEC 60335-2-41:2021 p.14 |
| **High** | 7.12.1 | Not Stated | Installation instructions must state maximum total head for pumps rated above 50 W. | Maximum total head for the 550 W pump is passed by reference to marking details, but no value is shown on the reproduced plate or elsewhere in the supplied record. | Provide the instructions and record the maximum total head in metres. | PDF pp.3,10 | Report: verified p.10 Standard: verified in IEC 60335-2-41:2021 p.14 |
| **High** | 15.1.2 | Not Stated | The supplied particular part specifies the pressure based on maximum depth or 15 m, if higher, for the 1.3-times branch. | The report records a 24 h immersion pass and a 1.3-times-pressure branch, but gives no maximum operating depth or actual test pressure; its alternative reference is 5 m rather than 15 m. | Establish the marked depth and document a correctly selected immersion pressure and test result. | PDF pp.4,17 | Report: verified p.17 Standard: verified in IEC 60335-2-41:2021 p.16 |
| **High** | 19.7 | Not Stated | The locked-rotor test is performed at rated voltage for the applicable period, with winding temperatures assessed accordingly. | The locked-rotor test records 116.6 V DC, inconsistent with the declared 220–240 V, 50 Hz rating; test duration is not recorded. | Repeat or substantiate the locked-rotor test at the verified rated supply and document its duration. | PDF pp.2,20–21,81 | Report: verified p.81 Standard: verified in IEC 60335-1:2010 p.54 |

| Priority | Clause | Gap type | Normative requirement | Finding reason | Required action | Pages | Evidence |
|---|---|---|---|---|---|---|---|
| **High** | 19.7; 24.5 | Not Stated | Motors with an auxiliary-winding capacitor require the specified locked-rotor capacitor fault tests, subject to the stated P2 exception. | Capacitor checks are marked P under 24.5, but the locked-rotor capacitor open- and short-circuit checks are marked N/A without explanation. | Establish whether an auxiliary-winding capacitor is fitted and document the applicable fault tests or exemption. | PDF pp.20,40 | Report: verified p.20 Standard: verified in IEC 60335-1:2010 p.54 |
| **High** | 24.1.4 | Not Stated | Applicable controls and motor protectors require evaluation under the stated component or Annex D provisions. | The particulars indicate a self-resetting thermal cut-out, while automatic-control and thermal-motor-protector assessments are marked N/A; the protector's role in compliance is not established. | Identify the cut-out and document its safety function and applicable evaluation. | PDF pp.4,38,62 | Report: verified p.4 Standard: verified in IEC 60335-1:2010 p.77 |
| **High** | 25.1; 25.3 | Not Stated | The particular requirements apply to submersible pumps other than class III. | The pump is declared class I and submersible, but the particular-part entries for a flexible cord and cord fitted with a plug are marked N/A; a general plug entry is passed. | Confirm and document the fitted flexible cord and plug for the tested pump. | PDF pp.4,40–41 | Report: verified p.40 Standard: verified in IEC 60335-2-41:2021 p.19 |
| **High** | 25.7; 25.8 | Not Stated | The supplied standards specify cord construction, conductor area and minimum length for applicable outdoor submersible pumps. | Outdoor use is indicated in the particulars, but cord type, conductor cross-sectional area and outdoor cord length are not established; the relevant entries are N/A. | Identify and verify the cord type, conductor area and length against the applicable requirements. | PDF pp.4,41–42 | Report: verified p.42 Standard: verified in IEC 60335-2-41:2021 p.19 |
| **High** | 25.10; 27.3 | Not Stated | Class I supply cords require a green/yellow earthing core, and the conductor arrangement must preserve earthing if the cord slips. | The report identifies a class I appliance with a supply cord, but protective-conductor colour and the conductor arrangement on cord displacement are marked N/A. | Inspect and record the protective conductor and its arrangement at the cord anchorage. | PDF pp.4,43,49 | Report: verified p.43 Standard: verified in IEC 60335-1:2010 p.82 |
| **High** | 27.5 | Not Stated | The standard requires a specified test current and a calculated resistance not exceeding 0.1 Ω. | Earth continuity is marked P, but no measured resistance, test current or tested connection is recorded. | Provide the earth-continuity test conditions and measured resistance for each applicable accessible metal part. | PDF p.49 | Report: verified p.49 Standard: verified in IEC 60335-1:2010 p.91 |
| **High** | 29.1; 29.2 | Not Stated | The insulation assessment must use the applicable rated impulse and working voltages, pollution degree and material group. | The clearance table uses a 500 V impulse rating and the basic-creepage table uses a working voltage of ≤50 V, despite the declared 220–240 V rating; basic clearance and creepage are reported as >2.0 mm. | Reassess and report each applicable insulation distance using the verified electrical ratings and construction. | PDF pp.2,51–54,83–85 | Report: verified p.83 Standard: verified in IEC 60335-1:2010 p.97 |
| **High** | 29.3; 25.13 | Not Stated | Supplementary insulation must be assessed by one of the specified routes. | The report passes the inlet bushing as supplementary insulation and states that insulation was checked by measurement, but gives no insulation thickness or alternative assessment result. | Provide the bushing insulation assessment and its measured thickness or applicable alternative test evidence. | PDF pp.43, 54–55,82 | Report: verified p.82 Standard: verified in IEC 60335-1:2010 p.101 |
| **High** | 30.1 | Not Stated | Relevant non-metallic and insulating parts require the heat-resistance assessment where deterioration could impair compliance. | The entire resistance-to-heat assessment is marked N/A and the appended table contains no part-specific test results or justification. | Identify the relevant parts and provide the required heat-resistance evidence or a justified exclusion for each. | PDF pp.55–56,87 | Report: verified p.55 Standard: verified in IEC 60335-1:2010 p.102 |
| **High** | 30.2; 30.2.1; 30.2.3 | Not Stated | The particular part selects 30.2.1 for the specified enclosed submersible-pump construction and RCD instruction, or 30.2.3 for other pumps. | Fire-resistance clauses and the appended table are marked N/A or blank, without establishing the construction and RCD-instruction conditions for the particular-part route. | Establish the applicable fire-test route and provide the relevant part or material evidence. | PDF pp.56–59,87 | Report: verified p.56 Standard: verified in IEC 60335-2-41:2021 p.20 |

| Priority | Clause | Gap type | Normative requirement | Finding reason | Required action | Pages | Evidence |
|---|---|---|---|---|---|---|---|
| **Medium** | — | Not Stated | The conflicting claim cannot be reconciled or evaluated against the unsupplied editions and amendments. | The test specification claims IEC 60335-2-41:2025, while the compliance summary claims IEC 60335-2-41:2012; the claimed Part 1 amendments and 2025 particular part were not supplied. | Provide the claimed standards and amendments and issue a report with a consistent test specification. | PDF pp.2–3 | Report: verified p.2 Standard: requirement not in supplied standards |
| **Medium** | — | Not Stated | The page's association with the report is uncertain, including its continuation of the creepage table. | The footer on PDF page 86 identifies report EN200622010S rather than EN201222034S. | Replace or verify the page under the correct report number. | PDF p.86 | Report: verified p.86 Standard: requirement not in supplied standards |